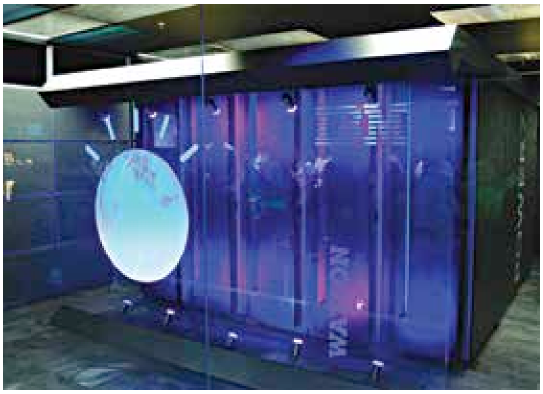
of supercomputers that are capable of deep-thinking. Tackling the problem of intelligence piece meal has already led to the creation of grandmaster-beating chess machines in the form of Watson and Deep Blue. However, when these titans of calculations are subjected to kindergarten level intelligence tests they fail miserably in factors of inferencing, intuition, instinct, common sense and applied knowledge. The ability to learn is still limited to their programming.

In contrast to these static computational supercomputers more organically designed technologies such as the delightful insect robotics are more hopeful. These "brains in a body" type of computers are built to interact with their surroundings and learn from experience as any biological organism would. By incorporating the ability to interface with a physical reality these applied artificial intelligences are capable of defining their own sense of understanding to the world. Similar in design to insects or small animals, these machines are conscious of their own physicality and have the programming that allows them to relate to their environment in real-time creating a sense of "experience" and the ability to negotiate with reality. A far better testament of intelligence than checkmating a grandmaster.

The largest pool of experiential data that any artificially created intelligent machine can easily access is in publicly available social media content. In this regard, Twitter has emerged a clear favourite with millions of distinct individuals and billions of lines of communications for a machine to process and infer. The Twitter-test of intelligence is perhaps more contemporarily relevant than the Turing Test where the very language of communication is not intelligently modern - since its greater than 140 characters.

The Twitter world is an ecosystems where individuals communicate in blurbs of thoughts and redactions of reason, the modern form of discourse, and it is here that the cutting edge social bots find greatest acceptance as human beings. These so-called socialbots have been let loose on the Twitterverse by researches leading to very intriguing results. The ease with which these programmed bots are able to construct a believable personal profile – including aspects like picture and gender – has even fooled Twitter's bot detection systems over 70 percent of the times.

The idea that we as a society so ingrained with digital communication and trusting of digital messages can be fooled, has lasting repercussions. Just within the Twitterverse, the trend of using an army of socialbots to create trending topics, biased opinions, fake support and the illusion of unified diversity can prove very dangerous. In large numbers these socialbots can be used to frame the public discourse on significant topics that are discussed on the digital realm. This phenomenon is known as "astroturfing" – taking its name from the famous fake grass used in sporting events - where the illusion of "grass-root" interest in a topic created by socialbots is taken to be a genuine reflection of the opinions of the population. Wars have started with much less stimulus. Just imagine socialbot powered SMS messages in India threatening certain communities and you get the idea.



How much compute power does functioning AI need?

But taking things one step further is the 2013 announcement by Facebook that seeks to combine the "deep thinking" and "deep learning" aspects of computers with Facebook's gigantic storehouse of over a billion individual's personal data. In effect looking beyond the "fooling" the humans approach and diving deep into "mimicking" the humans but in a prophetic kind of way - where a program might potentially even "understand" humans.

The program being developed by Facebook is humorously called DeepFace and is currently being touted for its revolutionary facial recognition technology. But its broader goal is to survey existing user accounts on the network to predict the user's future activity. By incorporating pattern recognition, user profile analysis, location services and other personal variables, DeepFace is intended to identify and assess the emotional, psychological and physical states of the users. By incorporating the ability to bridge the gap between quantified data and its personal implication, DeepFace could very well be considered a machine that is capable of empathy. But for now it'll probably just be used to spam users with more targeted ads.

From Syntax to Sentience

Artificial intelligence in all its current form is primitive at best. Simply a tool that can be controlled, directed and modified to do the bidding of its human controller. This inherent servitude is the exact opposite of the nature of intelligence, which in normal circumstances is curious, exploratory and downright contrarian. Man made AI of the early 21st century will forever be associated with this paradox and the term "artificial intelligence" will be nothing more than a oxymoron that we used to hide our own ineptitude.

The future of artificial intelligence can't be realised as a product of our technological need nor as the result of creation by us as a benevolent species. We as humans struggle to comprehend the reasons behind our own sentience, more often than not turning to the metaphysical for answers, we can't really expect sentience to be created at the hands of humanity. Computers of the future are surely to be exponentially faster than today, and it is reasonable to assume that the algorithms that determine their behaviour will also advance to unpredictable heights, but what can't be known is when, and if ever, will artificial intelligence attain sentience.

Just as complex proteins and intelligent life found its origins in the early pools of raw materials on Earth, artificial intelligence may too one day emerge out of the complex interconnected systems of networks that we have created. The spark that aligned chaotic proteins into harmonious DNA strands is perhaps the only thing that can possible evolve scattered silicon processors into a vibrant mind. A true artificial intelligence.